Math 55 quiz 01/27/2026

You must type or neatly handwrite each answer on a separate piece of paper and select the pages for each question when you upload to Gradescope. Feel free to use more than one sheet for an answer if needed. Note that both PDF and picture/photo format (JPG) are accepted. You must show your work. During the quiz, you may consult the textbook, your own notes and other course materials you yourself prepared before the beginning of the quiz. You may not consult resources online. You can post a private question to Ed Discussion to instructors only, but do not make public posts on Ed Discussion, Discord, or other online forums.

By submitting this quiz, you agree to adhere to the statement below: "As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others. All of the work submitted here is mine, and I have not discussed the content of this exam with any other person. I have not consulted any resources beyond the course materials and my notes during the exam. I have not used calculators, computers, or the internet to do calculations or look up answers."

1. Consider the two propositions:

$$\forall x \exists y\ (xy = 1), \qquad \exists x \forall y (x - y = 0)$$

Fill in the following table with the truth value of these propositions when x, y live in the following universes. Below the table, briefly justify your reasoning. By "positive" we mean $x, y > 0$.

	x, y are positive real numbers	x, y are positive integers
$\forall x \exists y\ (xy = 1)$		
$\exists x \forall y (x - y = 0)$		

Solution:

	x, y are positive real numbers	x, y are positive integers
$\forall x \exists y\ (xy = 1)$	T	F
$\exists x \forall y (x - y = 0)$	F	F

For the first row/proposition, note that we want to choose $y = \frac{1}{x}$. When x is a positive real number, so is y, so the first column is true. But when x is a positive integer, $y = \frac{1}{x}$ is not necessarily an integer. Hence the second column is false. For the second row/proposition, note there is not one x that works for every choice of y: we require $x = y$, but this depends on y. Hence both columns are false.

2. In the City of Knights and Knaves, you come across Alice, Beth, and Carla, standing in a queue, in that order, with Alice in front. (Remember: knights always tell the truth, and knaves always lie.) Each of them is either a knight or a knave.

- Alice says "There are at least three knights in this queue."
- Beth says "There are at least two knights in this queue."
- Carla says "There is at least one knight in this queue."

Prove that, if Alice or Beth is a knight, so is the person standing behind them. (There's nobody standing behind Carla, so you don't need to prove anything about her.) Therefore prove that, no matter how many knights and knaves there are in the queue, all the knaves are at the front and all the knights are at the back.

Solution: If Alice is a knight, then there are at least three knights in the queue, so there are at least two knights in the queue, so Beth is a knight. Similar logic holds for Beth and Carla.

Therefore, if anyone in the queue is a knight, so is everyone behind them, so any knaves must be right at the front.